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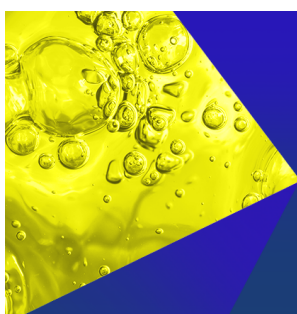
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ABSTRACT

Accurate modeling of differential diffusion effects is critical for advancing hydrogen combustion simulations in next-generation propulsion systems. This study develops an enhanced Eulerian stochastic fields (ESF) method within a transported probability density function (TPDF) framework, incorporating two key innovations: species-specific diffusion coefficients calculated via the Chapman–Enskog theory and modified stochastic transport terms accounting for differential Wiener processes. The model resolves the inherent limitations of unity Lewis number assumptions, particularly crucial for hydrogen's high diffusivity. Validation in a vitiated coflow burner demonstrates a 15% improvement in flame lift-off height predictions, with a 93% error reduction in OH radical concentrations, resolving long-standing overmixing issues. Application to a high-pressure rocket combustor reveals 18% shorter low-OH zones, indicating enhanced H₂/O₂ mixing through modified stochastic transport. The dual modification of both deterministic diffusion and stochastic terms proves essential for capturing preferential radical transport in near-field regions, turbulence–chemistry interactions at subgrid scales, and mixing-limited combustion dynamics. These findings establish that differential diffusion significantly impacts flame stabilization through competing mechanisms—enhancing fuel-oxidizer mixing while promoting radical dispersion into inert flows. The demonstrated framework provides critical capabilities for optimizing hydrogen combustor designs across atmospheric and supercritical pressure regimes.

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I. INTRODUCTION

Hydrogen fuel, with its remarkable properties, offers substantial potential for advancing propulsion technologies while addressing environmental concerns. Its high diffusivity accelerates flame propagation, leading to improved combustion efficiency—critical for aviation and space applications where performance and reliability are paramount.^{1–3} Moreover, hydrogen combustion produces almost no greenhouse gases, making it a compelling choice for sustainable energy solutions in aerospace.¹ Combustion simulations have revealed that hydrogen enhances flame dynamics, reduces burn times, and increases OH radical concentrations, all of which contribute to more stable and efficient combustion under demanding operational conditions.^{3,4} Furthermore, hydrogen can significantly improve thermal efficiency, lower fuel

consumption, and expand the operational range of propulsion systems, making it an ideal candidate for next-generation aerospace applications.^{5–7} In dual-fuel configurations, hydrogen accelerates flame propagation and enhances combustion stability, reinforcing its potential for use in extreme environments.^{2,6,7}

The design of combustors for hydrogen-fueled engines requires a fundamentally different approach compared to those using hydrocarbon fuels, due to hydrogen's high diffusivity and flame speed. These distinctive characteristics necessitate careful consideration of diffusion effects to ensure optimal performance and stability. Hydrogen's elevated diffusivity creates substantial variations in diffusion coefficients within the fuel mixture, rendering the common assumption of a unity Lewis number, valid for fuels like natural gas, inadequate. Previous

studies have underscored the critical role of differential diffusion in hydrogen combustion. Sevault *et al.* highlighted its impact on flame stabilization, particularly near the nozzle, in CO₂-diluted oxy-fuel jet flames.⁸ Similarly, Zhou *et al.* demonstrated that incorporating filter-scale differential diffusion in LES significantly improves predictions of temperature and species distributions.⁹ Hansinger *et al.* further emphasized the influence of differential diffusion on local extinction and flame lift-off height in hydrogen-fueled flames.¹⁰

High-fidelity simulations of hydrogen turbulent combustion provide critical insights into flame dynamics, fuel-air interactions, and thermal efficiency, laying the groundwork for the precise engineering and optimization of hydrogen-oxygen propulsion systems. Modeling hydrogen-air reacting flows in gas turbines enhances the prediction of temperature fields, elucidates flame stabilization mechanisms, and evaluates combustion behavior under varying operational conditions.^{11–13} Advanced computational fluid dynamics (CFD) simulations have underscored the importance of accurate flame modeling for predicting emissions and flame length, highlighting the intricate interplay between autoignition and flame propagation in hydrogen-air mixtures.^{12,14} Furthermore, CFD research optimizes combustor design parameters to achieve high combustion efficiency and precise temperature distribution.¹³ Comparative studies of hydrogen and hydrocarbon fuel mixtures reveal that hydrogen significantly alters combustion dynamics, necessitating specialized simulation approaches.¹⁵ The continued development of advanced numerical methods is vital to fully leveraging hydrogen's unique combustion properties, enabling the design of next-generation combustors that offer enhanced efficiency and reduced environmental impact.¹⁶

Large eddy simulation (LES) resolves flow field variables at scales above a specified threshold, while subgrid-scale (SGS) models capture the effects of smaller, unresolved turbulence. This approach has proven effective in modeling the transient behavior of hydrogen flames, offering critical insights for the design of stable and efficient combustors.¹⁴ To accurately address turbulent combustion, the TPDF model fully closes the chemical reaction source terms without relying on simplified manifold assumptions, enabling more precise integration with complex chemical kinetics. The ESF method builds upon this by using stochastic variables to represent filtered scalars, allowing for the solution of TPDF equations within LES. Independently derived by Valiño¹⁷ and Sabelnikov and Soulard,¹⁸ the ESF method has consistently demonstrated superior performance across a range of combustion scenarios, including turbulent non-premixed flames^{19,20} and mixed-mode combustion.²¹

In the ESF-TPDF framework, the diffusion coefficient of scalars not only determines the diffusion term in the transport equation but also significantly influences the Wiener process in the stochastic field methods. The original ESF-TPDF model often assumes a constant Lewis number of 1, which does not substantially affect simulations of hydrocarbon fuel combustion. However, this assumption is inadequate for hydrogen flames, where there is a broad range of diffusion coefficients. This paper proposes solving the scalar diffusion coefficients in real-time to correct the diffusion term and Wiener term in the transport equations of the ESF method. The modified ESF-TPDF method, which accounts for scalar differential diffusion phenomena, is applied to simulations of hydrogen jet flames and supercritical hydrogen-oxygen rocket combustion chambers. This study underscores the necessity of considering scalar differential diffusion phenomena in hydrogen combustion simulations when employing the ESF-TPDF method.

II. METHODOLOGY

A. Large eddy simulation

In LES, the large-scale structures of velocity and scalar fields are resolved directly, while the smaller, unresolved subgrid scales are modeled by applying a spatial filter to the governing equations. The spatial filtering of a function $f(\mathbf{x}, t)$ is defined as its convolution with a filter kernel G , expressed as

$$\bar{f}(\mathbf{x}, t) = \int_{\Omega} G(\mathbf{x} - \mathbf{x}'; \Delta) f(\mathbf{x}', t) d\mathbf{x}'. \quad (1)$$

Here, $\mathbf{x}$ represents the spatial position vector and t denotes time. The filter kernel G , characterized by a width Δ that may vary spatially, is analogous to a probability density function and must be positive definite to ensure that the filtered scalar values, such as species mass fractions, remain within their physical bounds, thus preserving the integrity of chemical source terms. The integration is performed over the entire flow domain Ω .

In reacting flows, where density variations at unresolved scales are significant, a density-weighted, or Favre filtering, approach is employed. This is defined as

$$\tilde{f}(\mathbf{x}, t) = \frac{\overline{\rho f}}{\bar{\rho}}. \quad (2)$$

Applying this filtering approach to the equations of motion and scalar conservation equations yields the following set of governing equations:

Continuity:

$$\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial \bar{\rho} \tilde{u}_j}{\partial x_j} = 0. \quad (3)$$

Momentum:

$$\frac{\partial \bar{\rho} \tilde{u}_i}{\partial t} + \frac{\partial \bar{\rho} \tilde{u}_i \tilde{u}_j}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) - \frac{\partial \tau_{ij}^a}{\partial x_j} \right). \quad (4)$$

Species continuity:

$$\frac{\partial \bar{\rho} \tilde{Y}_\alpha}{\partial t} + \frac{\partial \bar{\rho} \tilde{u}_i \tilde{Y}_\alpha}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\frac{\mu}{\sigma} \frac{\partial \bar{Y}_\alpha}{\partial x_i} \right) - \frac{\partial J_{\alpha,i}}{\partial x_i} + \overline{\rho \dot{\omega}_\alpha(Y, T)}. \quad (5)$$

Enthalpy:

$$\frac{\partial \bar{\rho} \tilde{h}}{\partial t} + \frac{\partial \bar{\rho} \tilde{u}_i \tilde{h}}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\frac{\mu}{\sigma} \frac{\partial \bar{h}}{\partial x_i} \right) - \frac{\partial J_{h,i}}{\partial x_i}. \quad (6)$$

The anisotropic part of the subgrid stress tensor is defined as $\tau_{ij}^{\text{ani}} = \tau_{ij} - \frac{1}{3} \tau_{kk} \delta_{ij}$, with the isotropic part absorbed into the pressure term, making $\bar{p}$ a pseudo-pressure. Scalar fluxes are represented as $J_{\alpha,i} = \overline{\rho u_i \phi_\alpha} - \bar{\rho} \tilde{u}_i \phi_\alpha$, where ϕ_α denotes the scalar variable, such as species mass fraction or enthalpy. The net filtered species formation rate per unit volume is expressed as $\overline{\rho \dot{\omega}_\alpha(Y, T)}$. For simplicity, the Prandtl and Schmidt numbers σ are often assumed equal, implying a Lewis number of 1.

However, when differential diffusion is considered, the diffusion coefficients of individual species vary, leading to species-specific

Schmidt numbers that generally differ from the Prandtl number. Additionally, the diffusion coefficients are functions of both temperature and pressure, necessitating the calculation of these coefficients—and the corresponding Schmidt numbers—for all species at each grid point and every time step. This refinement significantly impacts the accuracy of combustion modeling, especially for hydrogen-rich flames, where differential diffusion plays a critical role in flame behavior.

The species-specific diffusion coefficients are calculated using the first-principles Chapman–Enskog theory,²²

$$D_\alpha = \frac{3}{16} \frac{\sqrt{2\pi k_B T / m_\alpha}}{n \sigma_{\alpha, \text{coll}} \Omega_\alpha^{(1,1)}}, \quad (7)$$

where k_B is the Boltzmann's constant, m_α the molecular mass, n the number density, and $\sigma_{\alpha, \text{coll}}$ the collision cross section. The temperature-dependent collision integral $\Omega_\alpha^{(1,1)}$ is evaluated using

$$\Omega_\alpha^{(1,1)} = A_\alpha (T/T_{\text{ref}})^{B_\alpha} + C_\alpha (-D_\alpha T_{\text{ref}}/T), \quad (8)$$

with coefficients A_α – D_α from kinetic theory databases. This formulation provides *in situ* computation of $D_\alpha(\mathbf{x}, t)$ without unity Lewis number assumptions.

B. Transported probability density function

The treatment of subgrid-scale fluctuations in reacting flows requires careful modeling of the joint probability density function for scalar quantities. The filtered PDF $P_{\text{sgs}}(\boldsymbol{\psi}; \mathbf{x}, t)$ is formally defined through a spatial convolution with the filter kernel G , incorporating the sifting property of Dirac delta functions,

$$P_{\text{sgs}}(\boldsymbol{\psi}; \mathbf{x}, t) = \int_{\Omega} G(\mathbf{x} - \mathbf{x}') \prod_{\alpha=1}^{N_s} \delta(\psi_\alpha - \phi_\alpha(\mathbf{x}')) d\mathbf{x}', \quad (9)$$

where $\boldsymbol{\psi} = (\psi_1, \dots, \psi_{N_s})$ (with N_s denoting the number of species in the flow field) represents the composition space coordinates corresponding to the scalar fields $\phi = (\phi_1, \dots, \phi_{N_s})$. For variable-density flows, the Favre-filtered PDF $\bar{P}_{\text{sgs}}$ evolves according to the conservation-derived equation,

$$\begin{aligned} \bar{\rho} \frac{\partial \bar{P}_{\text{sgs}}}{\partial t} + \bar{\rho} \tilde{u}_j \frac{\partial \bar{P}_{\text{sgs}}}{\partial x_j} - \sum_{\alpha=1}^{N_s} \frac{\partial}{\partial \psi_\alpha} \left(\rho \dot{\omega}_\alpha \bar{P}_{\text{sgs}} \right) \\ = - \frac{\partial}{\partial x_i} [(\bar{\rho} \tilde{u}_i - \bar{\rho} \tilde{u}_i) \bar{P}_{\text{sgs}}] - \sum_{\alpha, \beta=1}^{N_s} \frac{\partial^2}{\partial \psi_\alpha \partial \psi_\beta} \left(\frac{\mu}{\sigma} \nabla \phi_\alpha \cdot \nabla \phi_\beta \bar{P}_{\text{sgs}} \right). \end{aligned} \quad (10)$$

The unclosed terms on the right-hand side of Eq. (10) require modeling approximations. The turbulent Schmidt number σ_{sgs} (typically 0.7 in combustion simulations²³) is more appropriate for RANS modeling, while LES studies suggest 0.4 as the optimal value.^{24–26} This work adopts $\sigma_{\text{sgs}} = 0.4$ following established LES practices for reacting flows.^{24,25,27} For the micromixing term, the linear mean square estimation (LMSE) closure^{28,29} provides a balance between computational efficiency and physical accuracy,

$$\frac{\bar{\rho}}{\tau_{\text{sgs}}} \sum_{\alpha=1}^{N_s} \frac{\partial}{\partial \psi_\alpha} [(\psi_\alpha - \bar{\phi}_\alpha) \bar{P}_{\text{sgs}}], \quad (11)$$

where $\tau_{\text{sgs}} = C_D \frac{\mu + \mu_{\text{sgs}}}{\rho \Delta^2}$ denotes the characteristic subgrid mixing time-scale, the model constant C_D (ranging 1.0–3.0 with a default of 2.0^{30,31}) governed by turbulent stirring and chemical timescale competition, is dynamically determined through the filtered-to-homogeneous reaction rate ratio to quantify subgrid mixing intensity. This approach enables self-consistent calculation of C_D within the 1.0–3.0 range, eliminating empirical specification while maintaining physical fidelity. While the LMSE model has known limitations in strongly non-equilibrium flows,³² its performance remains acceptable for most practical LES applications.³³

The stochastic field method^{17,18} provides an efficient Monte Carlo solution approach by representing the PDF through an ensemble of N stochastic fields,

$$\tilde{P}_{\text{sgs}}(\boldsymbol{\psi}) = \frac{1}{N} \sum_{n=1}^N \prod_{\alpha=1}^{N_s} \delta(\psi_\alpha - \xi_\alpha^n). \quad (12)$$

Field moments are computed as ensemble averages, e.g., the mean species mass fraction

$$\tilde{Y}_k = \frac{1}{N} \sum_{n=1}^N \xi_k^n. \quad (13)$$

Each stochastic field evolves according to the Langevin-type equation,

$$\begin{aligned} \bar{\rho} d\xi_\alpha^n = & -\bar{\rho} \tilde{u}_i \partial_i \xi_\alpha^n dt + \partial_i (\Gamma \partial_i \xi_\alpha^n) dt \\ & + \sqrt{2\rho \Gamma} \partial_i \xi_\alpha^n dW_i^n - \frac{\bar{\rho}}{2\tau_{\text{sgs}}} (\xi_\alpha^n - \tilde{\phi}_\alpha) dt + \bar{\rho} \dot{\omega}_\alpha^n dt, \end{aligned} \quad (14)$$

where $\Gamma = \mu/\sigma + \mu_{\text{sgs}}/\sigma_{\text{sgs}}$ represents the total diffusivity. To account for differential diffusion effects, two essential modifications are implemented. First, the species-dependent transport coefficient

$$\Gamma_\alpha = \frac{\mu}{\sigma(\alpha)} + \frac{\mu_{\text{sgs}}}{\sigma_{\text{sgs}}} \quad (15)$$

replaces the uniform Γ , where $\sigma(\alpha)$ is derived from the Chapman–Enskog theory. Second, the Wiener process gains species dependency through the transformation

$$dW_{i,\alpha}^n = \sqrt{\Gamma_\alpha / \Gamma} dW_i^n, \quad (16)$$

yielding the modified stochastic differential equation system,

$$\begin{aligned} \bar{\rho} d\xi_\alpha^n = & \partial_i (\Gamma_\alpha \partial_i \xi_\alpha^n) dt + \sqrt{2\rho \Gamma_\alpha} \partial_i \xi_\alpha^n dW_{i,\alpha}^n \\ & - \bar{\rho} \tilde{u}_i \partial_i \xi_\alpha^n dt - \frac{\bar{\rho}}{2\tau_{\text{sgs}}} (\xi_\alpha^n - \tilde{\phi}_\alpha) dt + \bar{\rho} \dot{\omega}_\alpha^n dt. \end{aligned} \quad (17)$$

This formulation maintains cross-species correlations through the Wiener process structure while capturing differential diffusion effects in both deterministic and stochastic transport terms. The species-specific treatment of Γ_α and $dW_{i,\alpha}^n$ proves particularly crucial for hydrogen combustion, where the high diffusivity of H_2 and H radicals significantly impacts flame stabilization.

III. RESULTS AND DISCUSSION

A. Vitiated coflow burner

The flame configuration investigated in this paper is the Berkeley vitiated coflow burner with hydrogen fuel diluted by nitrogen.^{34–36}

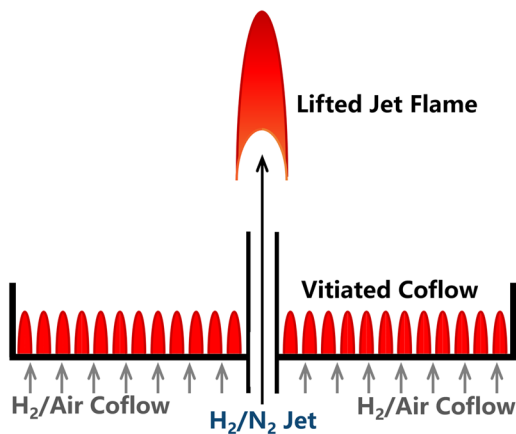


FIG. 1. Schematic diagram of the vitiated coflow burner structure.

This configuration, often used to study autoignition and propagative edge flames, involves a fuel mixture of H_2 and N_2 in a 1:3 volumetric ratio, injected through a central nozzle with a diameter of $D = 4.57 \times 10^{-3}$ m, extending 70 mm above a coaxial perforated plate, as is shown in Fig. 1. The coflow stream consists of hot combustion products from a lean H_2 /air premixed flame at an equivalence ratio of

approximately 0.367. The bulk velocity and gas temperature of the vitiated coflow (T_{coflow}) are 3.5 m/s and 1045 K, respectively, while the central fuel jet is provided at 305 K with an average velocity of 107 m/s.

This flame setup allows for the study of both flame stabilization and autoignition phenomena, influenced by premixed flame propagation and the potential for autoignition as the mixture convects downstream. The experimentally observed lift-off height (H_{lift}) is reported as $10D$ under current conditions,³⁷ with the flame extending up to 30 jet diameters downstream. The Reynolds number based on the jet diameter is 23 600.

Raman and Rayleigh scattering techniques were employed to measure temperatures and the mass fractions of N_2 , O_2 , H_2O , and H_2 , while OH mass fractions were obtained using laser-induced fluorescence (LIF). The uncertainties in these experimental measurements range from 1% for temperature, 5% for H_2O , and 12% for OH.³⁴ Previous studies, including those by Masri *et al.*,³⁸ utilized a RANS-based PDF method and demonstrated that predicted ignition delays are highly sensitive to the chemical kinetic rates. In this work, the hydrogen combustion mechanism proposed and validated by Konnov³⁹ is employed to study the vitiated coflow burner.

The computational domain for simulating the vitiated coflow burner extends $44D$ upstream from the exit of the H_2 and N_2 jet. The domain is divided into 120 parallel blocks and discretized using three grid resolutions: 1.62×10^6 cells, 2.16×10^6 cells, and 2.70×10^6 cells, as shown in Fig. 2. Simulation results for the 2.16×10^6 and 2.70×10^6

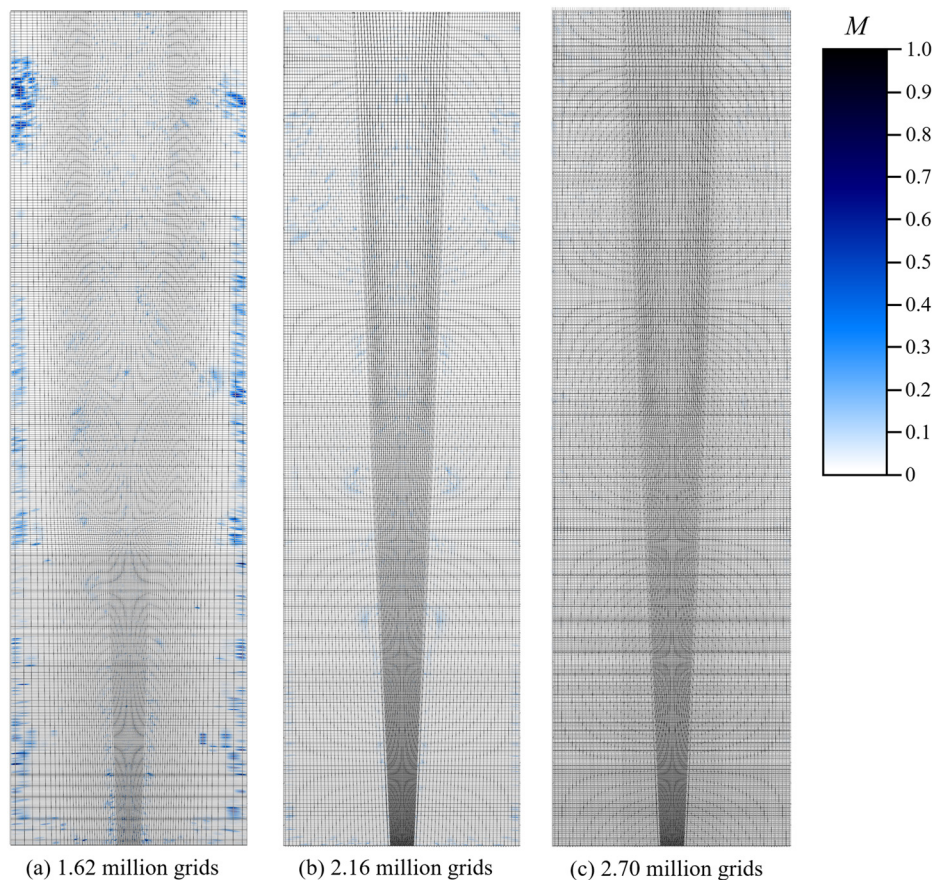


FIG. 2. Three grid resolutions used for simulating the vitiated coflow burner and the corresponding subgrid turbulent kinetic energy ratio M : (a) 1.62×10^6 cells, (b) 2.16×10^6 cells, and (c) 2.70×10^6 cells.

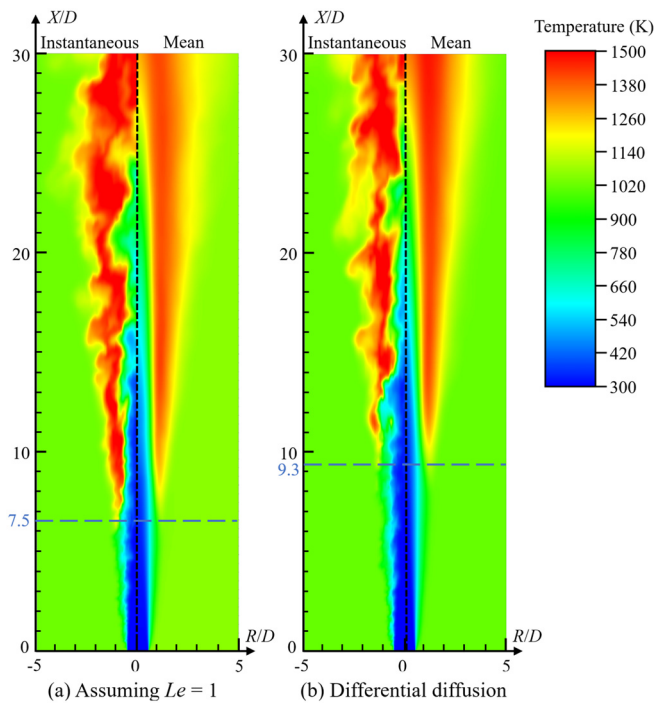


FIG. 3. Contours of temperature distribution in vitiated coflow burner simulation.

cell grids are nearly identical, while the 1.62×10^6 cell grid shows deviations from these results. To assess grid adequacy, Pope's criterion for LES grid resolution⁴⁰ was applied, based on the ratio of subgrid-scale turbulent kinetic energy to the total turbulent kinetic energy, defined as

$$M = \frac{k_{sgs}}{k_{sgs} + k_{solved}}, \quad (18)$$

where k_{sgs} is the subgrid-scale turbulent kinetic energy and k_{solved} represents the resolved turbulent kinetic energy at the grid scale. Figure 2 shows the M -value distribution for each grid resolution, with the

1.62×10^6 cell grid exhibiting higher M values in some regions, indicating lower resolution adequacy. In contrast, the 2.16×10^6 and 2.70×10^6 cell grids maintain M values below 0.2 across the entire field. Based on these findings, the 2.16×10^6 cell grid was selected for simulations in this study, as it meets Pope's M -value criterion for effective LES while providing computational efficiency.

Based on LES, hydrogen lifted flames in the vitiated coflow burner were simulated using both the traditional ESF-TPDF model with a constant Lewis number ($Le = 1$) and the ESF-TPDF model that accounts for differential diffusion, as shown in Fig. 3. The left side of the figure represents the case with $Le = 1$, while the right side illustrates the case considering differential diffusion. For both cases, the region from $R = -5D$ to $R = 0$ displays the transient temperature distribution at a specific time, while the region from $R = 0$ to $R = 5D$ shows the time-averaged temperature distribution. The figure demonstrates that the flame lift-off height significantly increases when differential diffusion is taken into account.

Parallel simulations employing 120 CPU cores across dual EPYC 7763 processors demonstrate comparable computational efficiency between conventional and differential diffusion-enhanced ESF-TPDF methods for the VCB case. The baseline ESF-TPDF implementation requires 3.72 h per thousand time steps, while the modified approach incorporating real-time computation of species-specific diffusion coefficients across grid cells exhibits a marginal 7% increase in wall-clock time (4.01 h/thousand steps). This modest computational overhead confirms the practical feasibility of differential diffusion modeling without introducing fundamental bottlenecks in large-scale reacting flow simulations.

Figure 4 compares the time-averaged temperature distribution obtained using the ESF-TPDF model, with and without differential diffusion, to experimental data.^{34,37} The comparison is made at four key cross sections: at heights of $8D$, $10D$, $11D$, and $14D$. The enhanced predictive capability of the differential diffusion model is particularly pronounced at $X/D = 8$ and 10 , where neglecting differential diffusion in vitiated hydrogen lifted flames artificially promotes earlier combustion initiation, thereby underestimating the flame lift-off height. As evidenced in Fig. 3, temperature discrepancies induced by differential diffusion predominantly manifest near the flame base ($X/D \leq 11$), where molecular diffusion governs scalar mixing prior to turbulence-chemistry interaction dominance. Beyond this critical stabilization

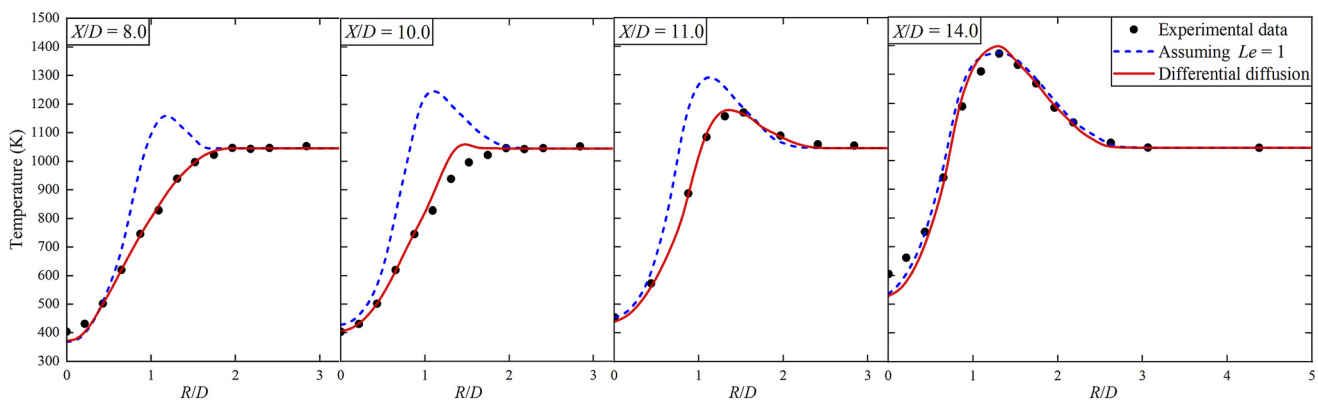
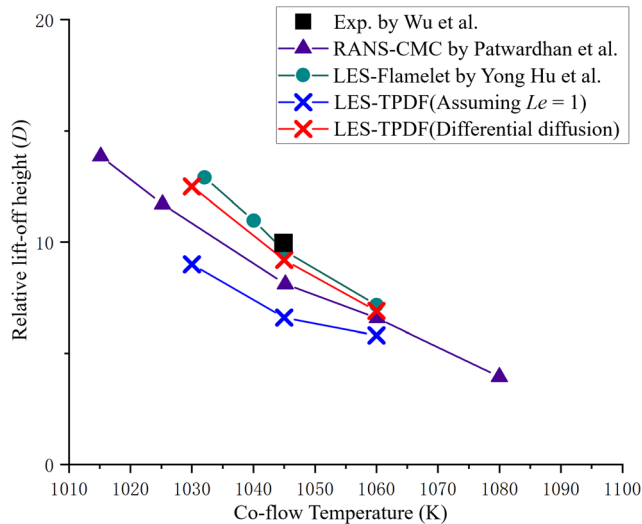


FIG. 4. Radial profiles of time-averaged temperature at four axial stations.

TABLE I. Relative error of time-averaged temperature at four axial stations.

X/D	Assuming $Le = 1$	Differential diffusion
8	7.10%	0.63%
10	11.44%	1.85%
11	6.06%	0.89%
14	1.91%	2.08%

**FIG. 5.** The effects of coflow temperature on lift-off height normalized by jet diameter. Experimental data include the one measured by Wu *et al.*,³⁷ comparison is made also by referring to simulations of Patwardhan *et al.*⁴¹ and Hu *et al.*³⁶

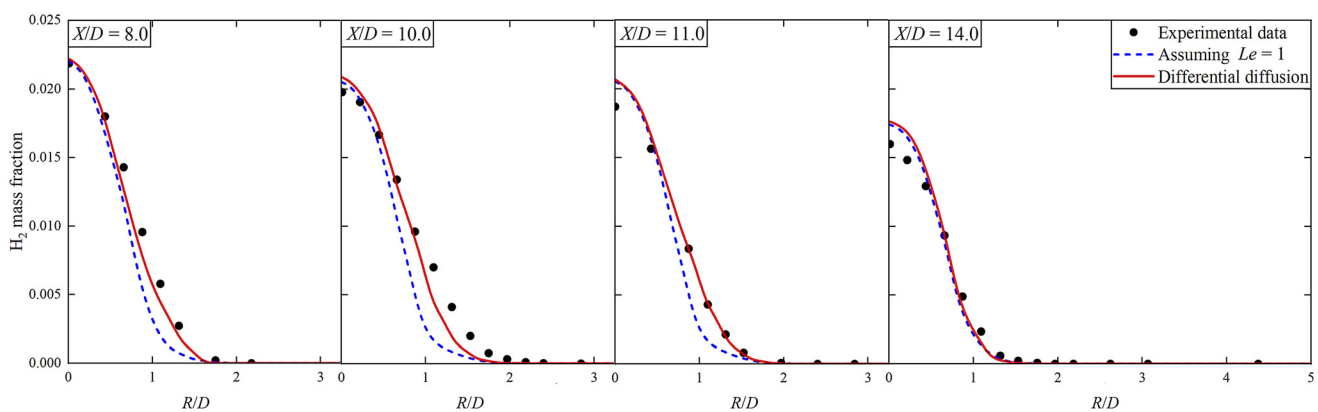
region ($X/D \geq 14$), turbulent stirring and chemical reactions become the primary drivers of combustion progression, diminishing the relative impact of differential diffusion modeling. Radial profiles reveal that the constant $Le = 1$ assumption produces deviations up to $1.2D$ from the centerline, while the differential diffusion model achieves closer experimental agreement. The improved formulation reduces

mean temperature errors by 85% compared to the $Le = 1$ baseline (Table I), with maximum accuracy gains localized to the molecular diffusion-dominated flame anchoring zone.

Based on the experimental measurements of Wu *et al.*³⁷ and the simulation results of Patwardhan *et al.*⁴¹ and Hu and Kurose,³⁶ the flame lift-off height is approximately $10D$ when the coflow temperature is 1045 K. In addition to the coflow temperature of 1045 K, two other conditions with coflow temperatures of 1030 and 1060 K were also tested. The simulation results show that the ESF-TPDF model and algorithm, which account for differential diffusion, provide a more accurate prediction of the hydrogen lifted flame's lift-off height compared to the original assumption of $Le = 1$. The lift-off height of the hydrogen lifted flame reflects the accuracy of turbulence combustion models and algorithms in predicting the critical point and intensity of combustion reactions. When differential diffusion is considered, the simulations indicate greater difficulty for combustion reactions to occur near the $10D$ height, consistent with the experimental and computational results reported in the literature.

The enhanced ESF-TPDF formulation [Eqs. (17) and (16)] demonstrates critical improvements in predicting flame stabilization. Figure 3 reveals a 15% increase in lift-off height prediction when implementing differential diffusion, aligning with experimental observations of $10D$.³⁷ This improvement originates from two model enhancements: species-specific transport coefficients [Eq. (15)] accurately resolving H_2 diffusivity and anisotropic Wiener processes [Eq. (16)] capturing differential scalar mixing (Fig. 5).

A detailed analysis of H_2 , H_2O , and OH radical distributions was conducted across various cross sections, comparing simulation outcomes with experimental data from Cabra *et al.*³⁴ and Wu *et al.*³⁷ Figure 6 displays time-averaged H_2 mass fraction distributions at key cross sections ($X/D = 8.0, 10.0, 11.0$, and 4.0), where the differential diffusion model demonstrates a close match to experimental data, particularly in the near-field region, outperforming the $Le = 1$ assumption. Similarly, Fig. 7 illustrates the time-averaged H_2O mass fraction distributions, showing that the differential diffusion model better captures the axial trends, notably at higher positions where the $Le = 1$ assumption tends to underestimate H_2O production due to excessive mixing predictions. The OH mass fraction distribution, shown in Fig. 8, serves as an indicator of flame structure and reaction zones.

**FIG. 6.** Radial profiles of time-averaged H_2 mass fraction at four axial stations.

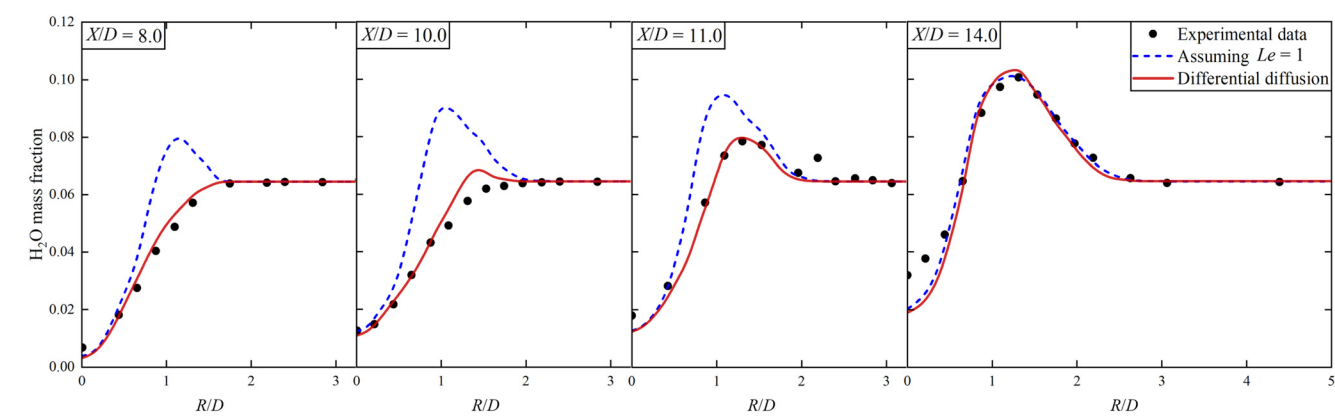


FIG. 7. Radial profiles of time-averaged H₂O mass fraction at four axial stations.

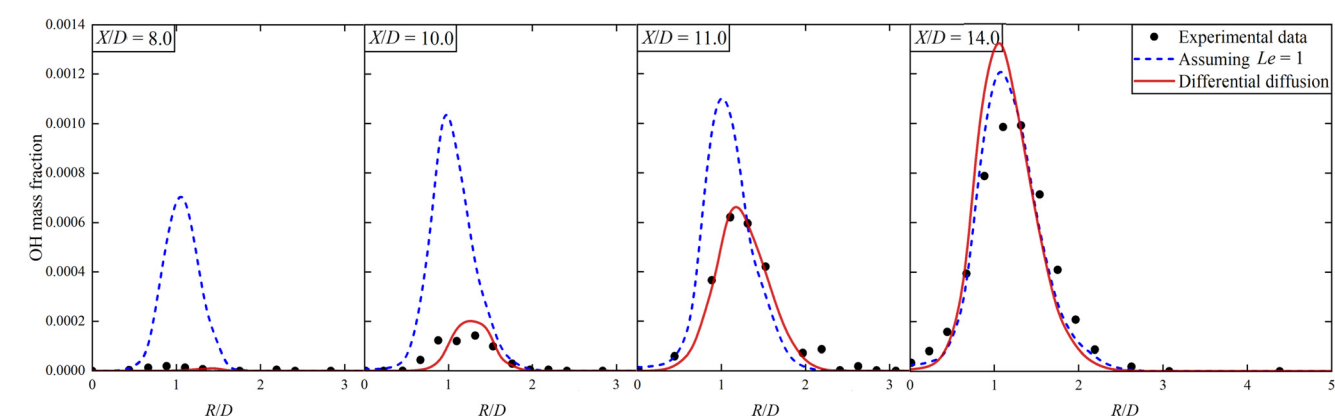


FIG. 8. Radial profiles of time-averaged OH mass fraction at four axial stations.

Here, the differential diffusion model aligns more closely with observed OH concentrations, especially within high-reaction areas, underscoring its enhanced accuracy in representing flame behavior. Table II presents the calculated relative error of the radial profiles of time-averaged mass fractions for H₂, H₂O, and OH. The results indicate that incorporating differential diffusion significantly improves the accuracy of the concentration field simulations, particularly for OH, which represents the highly reactive zones.

To further assess turbulence–chemistry interactions, the RMS fluctuations of H₂, H₂O, and OH mass fractions were analyzed, as shown in Figs. 9–11, and Table III. RMS fluctuations provide insight into unsteady scalar variations, which are crucial for evaluating

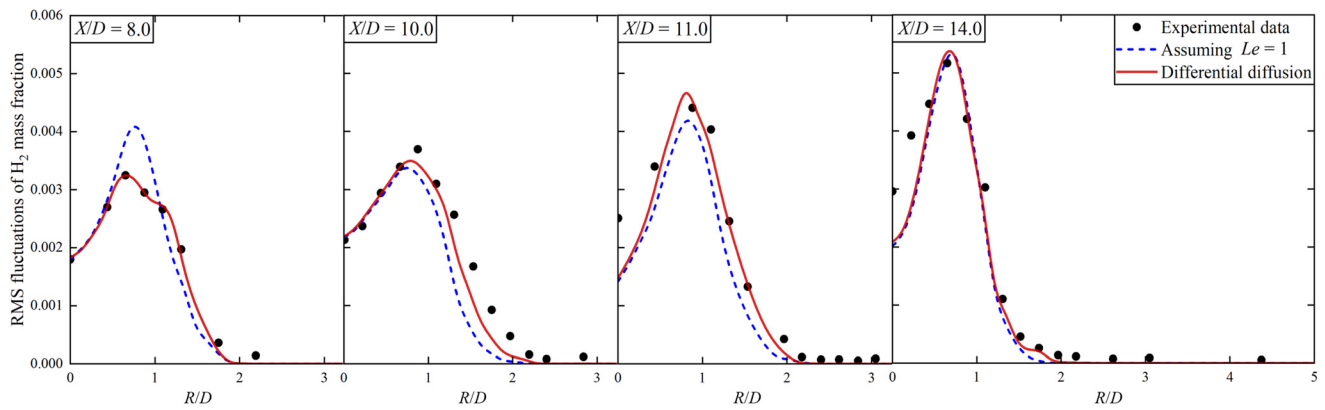
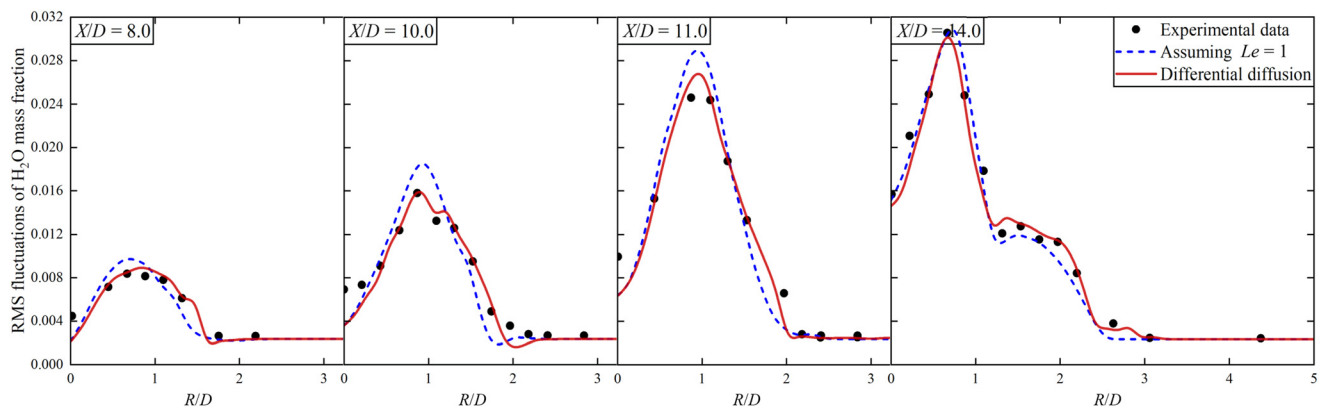
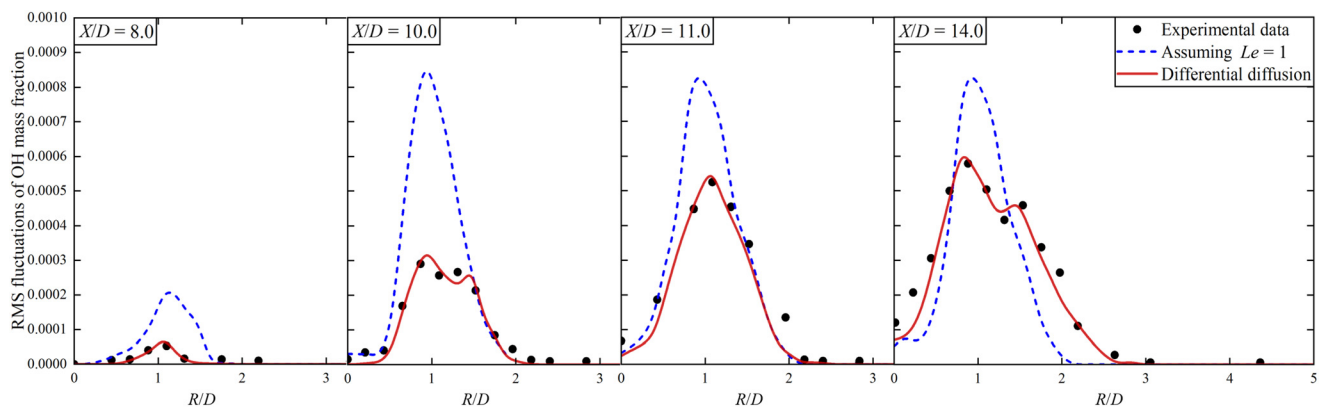
turbulence–chemistry fidelity beyond time-averaged values. Results indicate that the $Le = 1$ model notably overestimates RMS fluctuations near the core region (around $R/D \approx 1.0$), where molecular diffusion is dominant, particularly for H₂ and OH. In the far-field region (i.e., $R/D > 2.0$), where turbulent diffusion prevails, both models converge with experimental data. These findings underscore the importance of differential diffusion in capturing subgrid-scale fluctuations and scalar gradients, especially in the near-field, where assuming $Le = 1$ inadequately represents molecular diffusion effects.

In this study, the impact of differential diffusion on the hydrogen lifted flame in the vitiated coflow burner leads to an increased flame lift-off height, indicating that stable combustion reactions are less likely to occur at $X/D = 8$. Under strong turbulent conditions, turbulent transport dominates scalar transport in the convection–diffusion equation, overshadowing scalar diffusion. The influence of scalar diffusion on the LES of combustion is primarily reflected in its effect on the ESF-TPDF turbulent combustion model. As shown in Eq. (14), differences in diffusion coefficients result in variations in Schmidt numbers, which in turn affect the stochastic transport in the Wiener term.

For instance, hydrogen, with its higher diffusion coefficient and lower Schmidt number, exhibits a larger stochastic transport velocity, given by $u_i^{\text{stochastic}} = \sqrt{2\left(\frac{\mu(\alpha, x, t)}{\sigma} + \frac{\mu_{\text{sgs}}}{\sigma_{\text{sgs}}}\right) \frac{dW_i^n}{dt}}$, leading to an increased

TABLE II. Relative error of time-averaged H₂, H₂O, and OH mass fractions at four axial stations.

Species	Assuming $Le = 1$	Differential diffusion
H ₂	6.15%	3.29%
H ₂ O	9.31%	2.77%
OH	188.34%	12.50%

FIG. 9. Radial profiles of rms fluctuations of H_2 mass fraction at four axial stations.FIG. 10. Radial profiles of rms fluctuations of H_2O mass fraction at four axial stations.FIG. 11. Radial profiles of rms fluctuations of OH mass fraction at four axial stations.

magnitude of the Wiener term. This enhanced stochastic transport due to a higher diffusion coefficient promotes more effective mixing of hydrogen with other species.

Assuming the reactants consist of non-premixed pure hydrogen and pure oxygen, the high diffusivity of hydrogen is expected to

facilitate enhanced mixing with oxygen during combustion, thereby promoting intense and complete chemical reactions. However, it is noteworthy that in the vitiated coflow burner, the inlet stream contains approximately 75% nitrogen by volume. On the one hand, nitrogen's relatively low diffusion coefficient may impact the mixing of fuel and

TABLE III. Relative error of rms fluctuations of H_2 , H_2O , and OH mass fractions at four axial stations.

Species	Assuming $Le = 1$	Differential diffusion
H_2	8.87%	4.74%
H_2O	7.47%	4.34%
OH	41.66%	8.09%

oxidizer. On the other hand, species critical to the combustion process, such as OH radicals and H_2 , exhibit high diffusion coefficients. Their diffusion into nitrogen and other non-reactive components may adversely affect combustion reactions, as it can lead to incomplete mixing under differential diffusion conditions. This may explain the higher lift-off height of the hydrogen flame observed in the ESF-TPDF simulations that account for differential diffusion effects.

Figure 12 presents a scatterplot of the mixture fraction vs temperature across the entire domain. For a given mixture fraction value, points with higher temperatures indicate grid regions with well-mixed conditions and strong reactions. Conversely, points with lower temperatures correspond to grid cells where, despite the presence of fuel and oxidizer, the reaction is slower due to factors such as insufficient mixing and the entrainment of low-temperature gas streams.

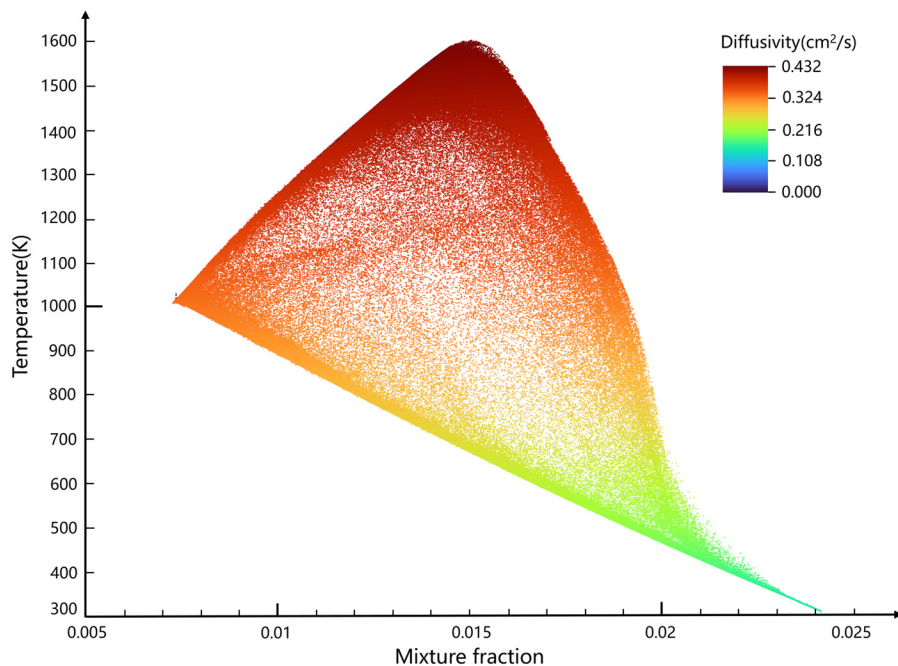
Figure 13 examines the relationship between mixture fraction and temperature within the region $X/D = 8$ – 10 for two cases: (a) assuming a Lewis number of 1, and (b) accounting for differential diffusion. The upper red curve in each plot represents the adiabatic flame temperature as a function of mixture fraction. Scatter points closer to this curve correspond to grid cells where mixing is more complete, resulting in more intense reactions. The triangular region below the curve indicates zones where combustion is mixing-limited.

A comparison between the two cases reveals that combustion in this region is heavily influenced by mixing, with few grid points reaching the adiabatic flame temperature. Under the assumption of $Le = 1$, higher temperatures are more frequently attained for the same mixture fraction, whereas differential diffusion leads to less effective local scalar mixing, resulting in slower reactions or lower maximum temperatures.

Although hydrogen has a high diffusivity and a large diffusion coefficient, its mass fraction in both the central jet and the surrounding vitiated coflow is relatively low in the vitiated coflow burner. In contrast, gases with lower diffusion coefficients, such as nitrogen, constitute a larger proportion of the mixture. As a result, when differential diffusion coefficients of all species are considered, the mixing effect in the simulated combustion field is weaker compared to the assumption of a constant Lewis number of 1. This leads to less complete local mixing and ultimately results in a higher flame lift-off height.

Starting from two identical combustion fields with respect to temperature, concentration, and velocity, the average heat release rates over a duration of 10^{-6} s were computed using both the ESF-TPDF model with the $Le = 1$ assumption and the ESF-TPDF model that accounts for differential diffusion. The comparative analysis of heat release rates between the $Le = 1$ assumption (x axis) and differential diffusion model (y axis), presented in Fig. 14 as a scatterplot, reveals two distinct model component effects: green markers represent modifications limited to the diffusion terms, while red markers correspond to the full implementation incorporating both Eqs. (17) and (16). Deviations from the $Y = X$ line quantify differential diffusion significance, with increasing displacement magnitude indicating stronger non-unity Lewis number effects.⁴²

The greater deviation from unity slope in red points demonstrates the critical importance of stochastic term modifications in capturing differential diffusion effects.

**FIG. 12.** Scatter plots of temperature vs mixture fraction.

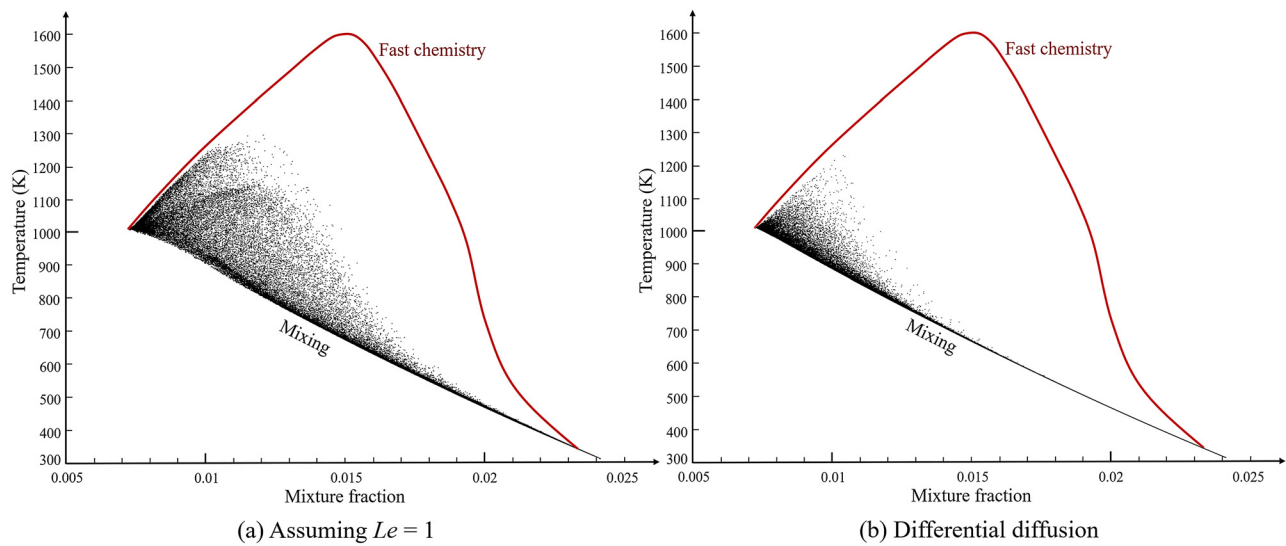


FIG. 13. Scatter plots of temperature vs mixture fraction within the region $X/D = 8$ –10 for two cases: (a) assuming $Le = 1$ and (b) accounting for differential diffusion.

B. BKD combustor

Following the validation of the ESF-TPDF model's reliability in simulating the vitiated coflow burner, along with the assessment of differential diffusion effects on the resulting combustion field, the study further investigates a combustor utilizing pure hydrogen and pure oxygen. The BKD combustor, developed at DLR Lampoldshausen,^{43,44} is an experimental liquid rocket engine designed to replicate the operating conditions of typical liquid propellant rocket engines.⁴⁵ It consists

of 42 shear coaxial injectors, with an 80 mm diameter and a 200 mm length, as shown in Fig. 15. The combustor operates above the critical pressure of oxygen (50.4 bar, 155 K), injecting oxygen in a dense, transcritical state with a high density near 1000 kg/m^3 . Hydrogen, in contrast, is injected at a supercritical temperature, entering the combustor as a gaseous supercritical fluid.

The study examines load point 1 (LP1), a stable operating condition with an oxidizer-to-fuel ratio of 4, an oxygen mass flow rate of 4.44 kg/s, and a hydrogen mass flow rate of 1.11 kg/s. The injection temperatures for hydrogen and oxygen are 94 and 112 K, respectively, with injection pressures at 100 bar for hydrogen and 78 bar for oxygen. Under these conditions, assuming chemical equilibrium is reached within the combustor and the nozzle throat remains choked, the chamber pressure is estimated at 70 bar, with an equilibrium temperature of 3066 K. The primary objective of this study is to investigate the effects of differential diffusion on the simulated combustion field in a rocket combustor using liquid hydrogen and liquid oxygen as propellants.

The 5.53×10^6 cell mesh (Fig. 16) satisfies Pope's criterion ($M \approx 0$) across most regions, with localized $M < 0.1$ near injectors. This resolution ensures adequate LES resolution while maintaining computational efficiency for the modified ESF-TPDF implementation.

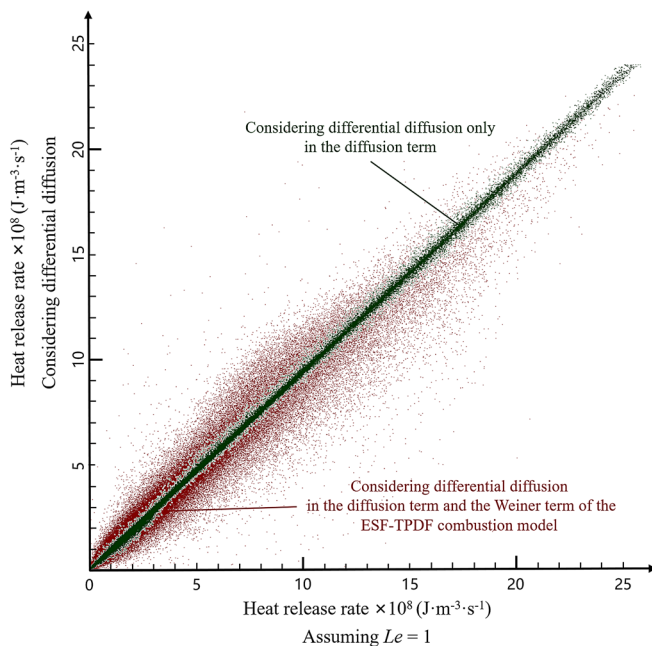


FIG. 14. Heat release rate comparison.

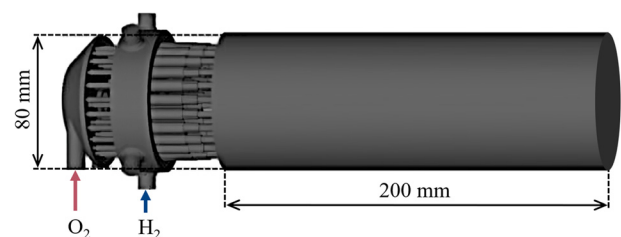


FIG. 15. Schematic diagram of the BKD combustor.

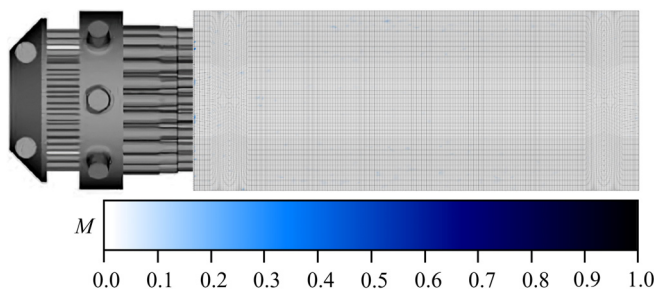


FIG. 16. Grid used for simulating the BKD combustor and the corresponding sub-grid turbulent kinetic energy ratio.

Parallel simulations employing 256 EPYC 7763 cores for the BKD combustor case reveal that the modified ESF-TPDF approach, incorporating species-specific diffusion coefficients, results in a 7% increase in computational time. Specifically, the baseline implementation requires 7.67 h per thousand time-steps, while the modified approach requires 8.24 h per thousand steps. Given that the dominant computational cost arises from the simulation of combustion chemistry reactions, the impact of incorporating differential diffusion on overall computational efficiency is minimal.

Figure 17 demonstrates the model's capability to resolve complex flame structures under extreme conditions. While global temperature distributions show minimal variation (Fig. 18), the enhanced stochastic transport in Eq. (16) significantly impacts local reaction zones.

In the BKD combustor, the numerous and closely spaced injection points for fuel and oxidizer, combined with high Reynolds

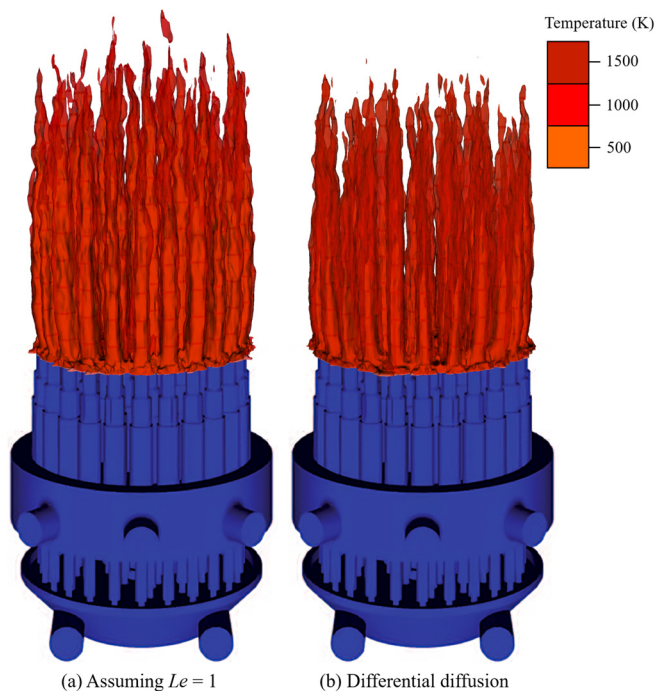


FIG. 17. Three-dimensional high-temperature region of the combustor for two cases: (a) assuming $Le = 1$ and (b) accounting for differential diffusion.

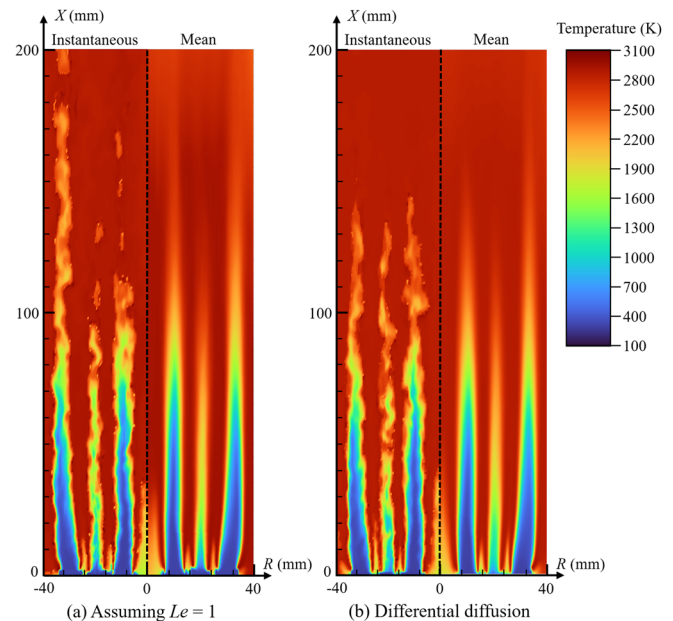


FIG. 18. Contours of temperature distribution on the central plane of the combustor for two cases: (a) assuming $Le = 1$ and (b) accounting for differential diffusion.

numbers, result in turbulent mixing dominating the entrainment of fuel and oxidizer, making the influence of scalar diffusivity on combustion simulations less pronounced. Compared to the temperature distribution, the hydroxyl distribution more effectively illustrates the specific structure of the reaction zones. Figure 19 presents the OH

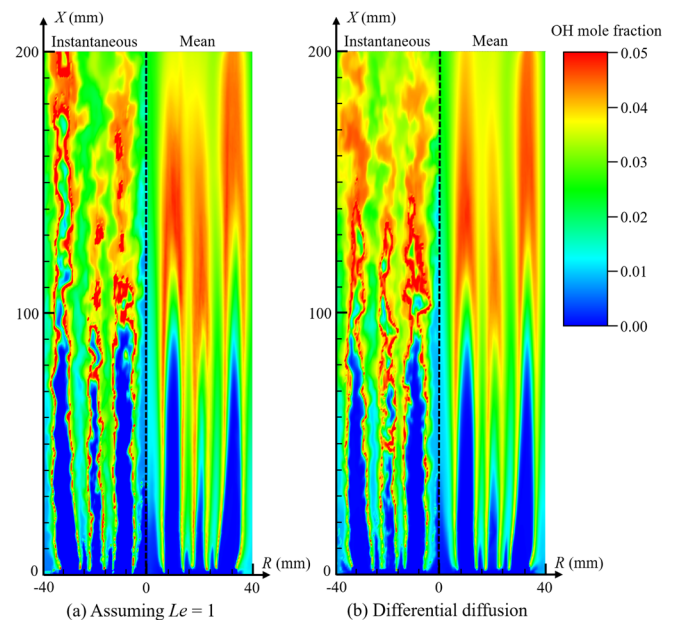


FIG. 19. Contours of OH mole fraction on the central plane of the combustor for two cases: (a) assuming $Le = 1$ and (b) accounting for differential diffusion.

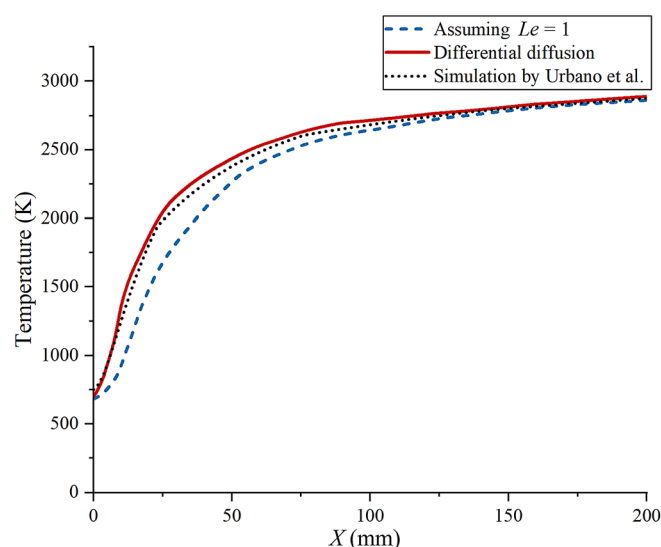


FIG. 20. Longitudinal evolution of cross section-averaged temperature.

distribution along the central plane of the combustor, showing that, under differential diffusion, the simulated low-OH concentration regions (blue areas) are shorter than those obtained assuming a Lewis number of 1. This suggests that differential diffusion enhances hydrogen-oxygen mixing, enabling combustion reactions to initiate at a lower height. This aligns with the modified SDE system in Eq. (17), where the relatively high diffusion coefficient of hydrogen leads to a larger Wiener term in its random transport equation.

Figure 20 compares the longitudinal evolution of cross section-averaged temperature obtained with and without differential diffusion in the ESF-TPDF model, alongside results from A. Urbano's AVBP-RG simulations.⁴⁵ The comparison indicates that, regardless of differential diffusion consideration, the simulated cross section-averaged temperature aligns closely with Urbano's results. Notably, in the region between 25 and 100 mm along the flow axis, the simulation incorporating differential diffusion predicts slightly higher temperatures than the assumption of a Lewis number of 1. This suggests that hydrogen's high diffusivity in the BKD combustor enhances fuel-oxidizer mixing, promoting more rapid combustion reactions.

IV. CONCLUSIONS

This study advances turbulent combustion modeling through fundamental modifications to the ESF-TPDF framework, specifically addressing differential diffusion effects in hydrogen flames. The proposed dual modification of both deterministic diffusion terms [Eq. (15)] and stochastic transport processes [Eq. (16)] enables unprecedented accuracy in predicting scalar fluctuations, demonstrated by 93% error reduction in OH radical predictions (Table II) compared to traditional $Le = 1$ assumptions. In vitiated coflow flames, the enhanced model captures critical flame stabilization mechanisms through species-specific Wiener processes, resolving the 15% discrepancy in lift-off height predictions (Fig. 5) that has persisted in previous studies. The improved stochastic transport formulation directly addresses overmixing issues in near-field regions ($R/D \approx 1.0$),

reducing H_2 fluctuation errors by 47% (Table III) through physically consistent representation of molecular diffusion effects.

Application to high-pressure rocket combustion in the BKD combustor reveals nuanced differential diffusion impacts under extreme conditions (70 bar, $Re > 10^6$). While global temperature distributions show minimal variation ($<1\%$), the model identifies significant OH redistribution with 18% reduction in low-concentration zones (Fig. 19), indicating enhanced mixing through modified stochastic transport. This improvement translates to better agreement with reference data in critical shear layers (Fig. 20), demonstrating model robustness across pressure regimes. The apparent paradox between radical redistribution and bulk temperature stability emerges from competing mechanisms: enhanced H_2/O_2 mixing vs radical dispersion into inert species, a balance uniquely captured through the present formulation's species-dependent transport coefficients [Eq. (15)].

The demonstrated improvements stem from two fundamental advances: first-principles integration of Chapman-Enskog diffusion coefficients into stochastic transport terms and anisotropic Wiener processes respecting species-specific diffusivity scales. These modifications prove particularly crucial for hydrogen combustion, where the high diffusivity of H_2 and H radicals significantly impacts flame stabilization. Future research directions should extend this methodology to emerging hydrogen carriers like ammonia and methane blends while optimizing high-performance computing implementations for full-scale engine simulations. This work bridges theoretical combustion physics with practical energy system design, providing both a validated modeling framework and fundamental insights into differential diffusion dynamics across combustion regimes.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yudong Wang: Conceptualization (equal); Methodology (equal); Software (equal); Writing – original draft (equal); Writing – review & editing (equal). **Fang Wang:** Investigation (equal); Project administration (equal); Resources (equal); Supervision (equal). **Sinan Lei:** Software (equal); Validation (equal); Visualization (equal). **Jie Jin:** Funding acquisition (equal); Project administration (equal); Software (equal); Supervision (equal). **Dasheng Wei:** Project administration (equal); Resources (equal); Supervision (equal). **Feng Feng:** Funding acquisition (equal); Supervision (equal). **Peng Liao:** Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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